

# Empirical Analysis of Risk for Sports Facilities

Neel Prajapati

*Dept. of Computer Science & Math*  
*Fairleigh Dickinson University*  
Teaneck, USA  
n.prajapati@student.fdu.edu

Adarsh Dhorajiya

*Dept. of Computer Science & Math*  
*Fairleigh Dickinson University*  
Teaneck, USA  
a.dhorajiya1@student.fdu.edu

Burhan Petiwala

*Dept. of Computer Science & Math*  
*Fairleigh Dickinson University*  
Teaneck, USA  
b.petiwala@student.fdu.edu

Avimanyou Vatsa

*Dept. of Computer Science & Math*  
*Fairleigh Dickinson University*  
Teaneck, USA  
avatsa@fdu.edu

Alexander Casti

*Dept. of Computer Science & Math*  
*Fairleigh Dickinson University*  
Teaneck, USA  
casti@fdu.edu

**Abstract**—In today’s digital landscape, organizations depend on intricate information systems to manage sensitive data, maintain operations, and ensure business continuity. As these systems become more complex, the need for effective risk management strategies grows. Improving facilities, student satisfaction, event attendance, and sustainability are great successes in operational policies, procedures, and event management. Additionally, the sports building serves a diverse community, including students, faculty, staff, and external guests, supporting daily recreation, varsity athletics, and large-scale events. Therefore, this study presents a focused risk management approach based on the NIST Cybersecurity Framework (CSF 2.0) for athletics centers in sports facilities, such as indoor game spaces, fitness centers, and outdoor game spaces, at academic institutions. Thus, this study aims to evaluate asset management practices and identify physical and logical vulnerabilities by conducting in-depth interviews with students, coaches, and other supporting staff, and by gathering firsthand insights into operational workflows, physical security measures, and system-level safeguards. These qualitative findings, combined with realistic frameworks, form the foundation of analysis. In conclusion, sports operations and this study underscore the institution’s commitment to technical reliability, risk awareness, and corresponding mitigation, as well as 24/7 assistant services.

**Index Terms**—NIST, Cybersecurity Framework, Athletic Center, Risk Management, Risk Analysis, Gap Analysis.

## I. INTRODUCTION

In increasingly connected campus environments, including athletic centers and facilities, sports fitness centers, and other sports activity areas. These are no longer just physical spaces but integrated ecosystems combining technology, health, and user interaction. These places exemplify this shift, employing biometric access systems, IoT-enabled fitness equipment, and digital member management platforms to enhance user convenience and operational efficiency. However, with this integration comes a growing array of cyber risks. These include cyber threats, unauthorized access, system downtimes, and personal data breaches. As a result, securing the athletic center’s infrastructure is essential to safeguard sensitive information, ensure uninterrupted services, and protect users. The center’s ecosystem includes biometric scanners, surveillance,

and digital health records—each carrying potential cyber risks [8].

Additionally, the risk management at the athletic centers requires careful attention. In particular, the digital transformation of academic sports buildings has expanded their role beyond physical training facilities into innovative environments that integrate technologies such as biometric access systems, digital scoreboards, IoT-enabled fitness equipment, and online ticketing platforms. Thus, its critical digital assets—such as CCTV surveillance, Wi-Fi networks, athlete performance databases, and payment platforms—require robust protection against threats and vulnerabilities. While these innovations enhance operational efficiency and user experience, they also introduce significant cybersecurity vulnerabilities [4], [5].

Implementing strong cybersecurity measures at the athlete Center is crucial for enhancing business value. Robust data encryption and access controls improve trust by protecting sensitive data and mitigate legal risk by ensuring regulatory compliance. This also increases reliability for systems like athlete performance databases. Investing in a Security Information and Event Management (SIEM) solution increases reliability through better threat detection, minimizing disruptions to events. It also helps mitigate legal risk and builds trust by improving incident response. Supplier risk assessments are essential for legal risk mitigation, ensuring vendor security practices align with the center’s standards. This also provides Financial Returns by preventing costly security incidents and enhancing Reliability, along with effective IT asset management, which improves reliability by ensuring accountability for hardware and software. It also leads to financial returns through cost optimization. Cybersecurity training enhances Reliability by reducing human error and fosters trust [9], [14]–[16].

Therefore, this study conducts a targeted risk management of sports facilities, associated assets, and personnel. It focuses on identifying physical, digital, and administrative vulnerabilities and threats. The NIST Cybersecurity Framework (CSF 2.0) serves as a standard that defines five major assessment

functions: Identify, Detect, Protect, Respond, and Recover [10], [17], [18]. The study assesses current practices, visualizes risk gaps, and proposes actionable strategies to reduce exposure and enhance operational resilience. This framework-based analysis identifies and evaluates risks affecting a university-operated fitness center's operations and information systems. Furthermore, it evaluates and strengthens the cybersecurity posture of a university sports building using the CSF 2.0, which provides structured guidance across five core functions: Identify, Protect, Detect, Respond, and Recover. In summary, the findings reveal strengths in monitoring and asset control and critical gaps in incident mitigation and recovery readiness. Recommended strategies include formalizing policies, enhancing vendor management, and integrating AI-based detection tools to improve resilience and continuity [19], [20].

## II. BACKGROUND

With technological advancements, however, comes a critical need to ensure that systems and data are protected from cyber threats. Potential vulnerabilities include unauthorized physical or digital access, data breaches involving personal health information (PHI), poorly secured network endpoints, and reliance on third-party vendors without proper vetting. These challenges are not hypothetical—they pose real threats to user privacy, operational continuity, and regulatory compliance [21], [22].

As cybersecurity is a growing field, many unaccounted-for vulnerabilities can cause users to lose control of their systems. Many different types of entities are affected by these cyber risks, including individuals, corporations, and governments. Some effects of these exposed vulnerabilities include financial loss, data theft, and negative changes to an individual's reputation or a company's brand. With an ever-growing network of connections, more vulnerabilities are discovered every day. Therefore, a public entity such as the CVE, which is sponsored by the U.S. Department of Homeland Security, reports on these common vulnerabilities. Their datasets report all currently known vulnerabilities, including their severity, the affected systems and software, and potential mitigation strategies. With this amount of data, performing research through risk analysis might become useful [1], [2].

Moreover, the cybersecurity Risk Quantification and Classification Framework. It focuses on two main points — likelihood (how likely a breach will happen) and impact (how much damage it will cause). The study used real data from S&P 500 companies, collected from the Privacy Rights Clearinghouse database. They examined various types of data breaches, including hacking, insider misuse, data theft, lost devices, and unauthorized access. After studying all this, they used a Likelihood-Impact Matrix to assess the risk of each breach type. Their research showed that hacking and data loss from stolen devices are the most dangerous and most common breach types. These incidents usually cause the biggest financial loss and legal problems. In citezadeh2023, the authors also used content analysis to examine how companies respond to data breaches and how long it takes them to recover.

They added data visualization dashboards to make the results easy to understand for business leaders [11], [23]–[25].

Many security model considers all layers of the ecosystem, including perception protocols such as Zigbee, Wi-Fi, and 6LoWPAN, as well as communication protocols such as HL7, HTTP, MQTT, and CoAP. It also analyzes identity mechanisms such as SAML and OAuth to understand how identity providers and service providers interact. Through detailed case modeling, the authors show how weak implementations of SAML or OAuth can allow token theft, malicious redirects, and poor session handling. Similarly, they highlight vulnerabilities in IoMT devices, such as wireless exploits, weak join keys, and improper authentication, which can let attackers move from one compromised device to the wider network. [12], [26], [27]

Each threat is mapped in an attack tree and assigned a risk score, helping identify which components need the most immediate attention. In [12], the authors recommend a mix of administrative and technical controls. Administrative measures include governance, configuration management, and proper training, while technical measures cover endpoint security, segmentation, intrusion detection, encryption, and multi-factor authentication. For identity management, they suggest improving SAML and OAuth implementations, enforcing strict code reviews, and even exploring blockchain-based identity systems to strengthen trust and privacy. [12], [28].

The study uses a qualitative grounded theory approach with semi-structured interviews of twelve security leaders from major United States professional leagues and cybersecurity experts. Participants discuss threats across three areas: physical security, personnel security, and cyber security. On the physical side, the picture is relatively strong. Stadiums have invested heavily in perimeter control, screening, cameras, bollards, and visible law enforcement, guided by national bodies and joint work with agencies such as the Department of Homeland Security. Fans increasingly expect to see these measures and even view them as part of the event experience.

Personnel security of sports personnel is also fairly mature. Facility managers rely on partnerships with police and other public agencies, clear codes of conduct for spectators, and trained security staff who monitor crowds, share intelligence, and respond to disorder or possible attacks. The interviews highlight the importance of information sharing between private venue operators and public authorities, especially when dealing with insider threats or coordinated attacks around large events [13].

## III. MATERIALS AND METHODS

The qualitative data are collected through internal reviews and staff interviews at a few academic athletic facilities. Sports personnel, including sports persons and students, are identified and scheduled for interviews. During the interview, a reference sheet is provided to interviewees so they can answer accurately and appropriately about qualitative and quantitative values. A set of 160 questions is prepared to interview athletic center personnel to ensure compliance with CSF 2.0's five activities:

Identify, Protect, Detect, Respond, and Recover. The collected datasets were processed for the risk analysis method. During the interviews, many questions went unanswered. Thus, this approach excludes those unanswered questions and uses pre-processed data for the analysis.

The risk assessment is adopted from the CSF 2.0, focusing on five core domains: Identify, Protect, Detect, Respond, and Recover. The collected and preprocessed data are used to find the average in each category and subcategory of risk framework activities. To evaluate the implementation status of 23 cybersecurity activities, each activity was scored based on perceived effectiveness and alignment with best practices. Average scores are compared to predefined target benchmarks, and performance gaps are used to identify critical risk areas requiring immediate attention. After that, the top five gaps were identified and addressed with appropriate mitigation measures [10].

#### IV. RESULTS AND OBSERVATIONS

Security gaps were not limited to low-scoring activities; they also occurred in well-rated areas due to inconsistent application, weak documentation, or informal processes. The mitigation plans should focus on staff training, clearer communication protocols, and AI-based tools to enhance threat detection. Prioritizing these gaps in the next phase of implementation can improve business continuity.

##### A. Case Study of Athletic Center

The process began by identifying key digital assets and services within the facility, including biometric access systems, CCTV surveillance, IoT gym equipment, digital scoreboards, and online ticketing platforms. Stakeholder engagement is central to the approach, involving structured interviews with four individuals working across IT support, security, operations, and third-party maintenance roles. These interviews help gather insights into current cybersecurity practices and challenges.

A risk management questionnaire aligned with the five CSF functions—Identify, Protect, Detect, Respond, and Recover—is used to evaluate areas such as asset inventory, access controls, incident handling, and data backup procedures. The responses were scored to generate an “Actual” score for each function, which was then compared against a “Target” score of 6.0. The results, compiled and averaged from all five participants, revealed varying gaps in each function, as visualized in the results table. These findings served as the basis for identifying vulnerabilities and creating recommendations to improve the facility’s cybersecurity resilience.

The cybersecurity assessment of the athlete Center reveals several critical gaps that require targeted mitigation strategies to enhance the facility’s overall security posture. The top five gaps indicate areas where current practices fall short of the target performance level. These gaps span multiple cybersecurity functions and require a prioritized, comprehensive remediation approach. To effectively address these weaknesses, specific mitigation strategies must be implemented. For instance, to

improve Supply Chain Risk Management (ID.SC), a robust vendor risk management program should be established, incorporating security assessments, contractual obligations, and regular audits of third-party suppliers. Asset Management (ID.AM) can be strengthened by deploying automated asset discovery tools, implementing asset classification systems, and establishing secure asset lifecycle management procedures. Enhancing Awareness and Training (PR.AT) requires regular cybersecurity education for all personnel, simulated phishing exercises, and role-based training programs tailored to different user groups. Data Security (PR.DS) can be improved by implementing multi-factor authentication, data encryption, and data loss prevention mechanisms. Finally, addressing the largest gap, Anomalies and Events (DE.AE), necessitates the deployment of intrusion detection and prevention systems (IDPS), security information and event management (SIEM) systems, and the establishment of rigorous security log monitoring processes.

In summary, the top five security gaps found in athletic center facilities through qualitative interviews of sports personnel. These are Supply Chain Risk Management, Asset Management, Awareness and Training, Data Security, and Anomalies and Events.

##### B. Case Study of a Fitness Center or Recreation Center

Evaluating risk involves two phases: gap analysis and risk treatment. The gap analysis includes data acquisition through interviews and the identification of gaps among the primary NIST cybersecurity framework functions (CSF 2.0). These functions contained several sub-functions (e.g., Identify included ID.AM-1, ID.AM-2, etc.) used to compute the gap concerning the target value (6.00). The risk treatment phase analyzes, mitigates, and finds significant business impact on computing services. The computing services’ impacts are concluded based on Gaps and corresponding mitigations, as determined by the center’s needs. Also, an internal marketing and communication plan has been proposed using existing channels to ensure awareness and adoption. Each mitigation effort is tied to a key business benefit, such as technical risk mitigation, financial returns, legal risk mitigation, or increased operational reliability.

In summary, the top 5 security gaps found in a fitness or recreation center through qualitative interviews of participants.

- Continuous monitoring of surveillance cameras.
- The access control mechanism may be improved.
- The response rate from security facilities needs to be improved.
- The risk management strategies need to be redefined and implemented over time.
- A continuous risk assessment must be done by risk management professionals.

#### V. CONCLUSION

To assess risks, structured interviews are conducted with sports staff and students, security personnel, and operations managers, complemented by a NIST CSF-based questionnaire

Key findings revealed, which includes security gaps (Limited multi-factor authentication (MFA) on admin systems, outdated access logs, and lack of network segmentation), operational weaknesses (No formal backup testing or change management protocols), and response challenges (Inconsistent communication during incidents across stakeholder roles). This study identified five major security gaps, corresponding mitigations, and their impact on event management and the security of athletes and assets.

Therefore, this study aims to evaluate the effectiveness of existing controls and offer actionable recommendations to strengthen the athlete and recreation or fitness center's cybersecurity posture on IT Risk Management, featuring scored responses to standardized cybersecurity questions.

#### ACKNOWLEDGMENT

We gratefully acknowledge the support of the Research-release time, collaborators, and other lab members of the Deep Chain (DC) lab at the Olsen College of Engineering & Science (OCES), Fairleigh Dickinson University (FDU), Teaneck, New Jersey 07666.

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